

Space age

We look at planets that exist where none should have. Read on to find out why they're so mysterious.

**Sea levels**

Measuring sea levels is a lot more complex than it seems. Read on to know how: http://digit.in/c_lvl

A Geneteel way of doing things

Why gene therapy could be the answer to all of the world's biggest and baddest diseases and why progress has been so slow, so far.

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"There is a killer inside you. You don't have to deny it. We were created to be that way."

While Liquid Snake's 'grand' speech to Solid Snake in the finale of the PS1 game, highlighted a convoluted story, Metal Gear's projection of the possibilities of gene therapy were admittedly not based on as much hogwash as it would first seem.

What exactly is gene therapy?

Genes contain the recipes required to make proteins which your body needs for energy and repair and maintenance, among other things. If a gene goes bad, as expected, the proteins too get messed up. You solve this problem by adding a nice new copy of that gene so that it can make the protein is how it is supposed to be. That is gene therapy. One of its biggest advantages over pumping the patient with drugs is that, in this case, the underlying base cause of the patient's problems is removed.

How do you treat an ill gene?

Well, that depends on how serious the problem is. Picture a professor who purposely teaches his students incorrectly because of which his students have all their concepts muddled up and don't know why he keeps flip-flopping. That is a problem. So, we introduce an interesting professor who really knows his subject.

In the first scenario, the new professor (the good gene) begins teaching



Before nanomachines, so many of these series were obsessed with genes

the students and slowly but surely the confusion of the students dies. Here what has happened is that a new gene has been merely added to the body, which takes over the protein which the defective gene was responsible for and helps create the protein as it as it should be.

In the second scenario, our evil professor (the malfunctioning gene) walks into the class while the new professor is teaching and stands in the centre of the room and begins lecturing very loudly. Nobody is able to hear our nice professor over the evil professor and so the students continue to remain confused about the subject. That professor is what is called a 'dominant negative'. Here merely adding a working gene will not be enough. This has to be cured by either repairing the defective gene, or replacing it entirely. (In this case by getting rid of the evil professor who was deliberately injecting the class with knowledge that was not only incorrect but was also responsible for plenty of confusion and disarray).

How is this gene inserted?

By fighting fire with fire. What may surprise you is that one of the preferred vehicles for a new gene to be transplanted in the body is a virus. Yes the same kind which causes HIV/AIDS. But why? Because the damned things are good at what they do – they are good at targeting and entering cells. And when it comes to kicking asses of diseases, scientists hold no grudges.

Take Professor Harrich from Queensland Institute of Medical Research for instance. He developed a way to use HIV to beat HIV, although it is still in the initial stages. By creating the 'Nullbasic' protein by mutating an existing HIV protein through genes, Harrich was able to stop the spread of the HIV virus.

So...Supermen?

Just go ahead and say it. Can we make super-people?

Well not as of yet. Super-babies? Probably (very, very 'probably'). See, to